

EIELSON AFB, AK, USA

WMO: 702650

Lat: 64.683N

Lon: 147.083W

Elev: 167

StdP: 99.34

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26407

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.9	(c) -38.3	(d) -40.0	(e) 0.1	(f) -35.6	(g) -37.3	(h) 0.1	(i) -33.6	(j) 8.0	(k) -1.5	(l) 6.8	(m) -5.8	(n) 0.6	(o) 170	(p) 0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	27.2	15.9	25.2	15.6	23.5	14.8	17.6	24.7	16.6	22.9	15.7	21.4	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.9	19.0	14.0	10.2	17.9	13.0	9.5	17.4	50.0	24.9	46.8	23.0	44.3	21.6	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.1	5.2	-42.3	30.5	3.5	1.8	-44.8	31.8	-46.8	32.8	-48.8	33.8	-51.4	35.1
			-38.5	19.4	3.8	1.3	-41.2	20.4	-43.5	21.1	-45.6	21.8	-48.4	22.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-21.6	-16.6	-10.5	1.2	9.7	15.1	16.5	13.3	7.5	-4.3	-15.9	-20.9		
	DBStd	15.42	10.14	8.29	7.85	5.94	4.17	3.21	2.62	3.17	4.10	6.63	8.20	8.86		
	HDD10.0	4959	981	744	636	266	56	1	0	6	93	443	776	957		
	HDD18.3	7497	1239	977	894	516	268	106	68	158	326	702	1026	1216		
	CDD10.0	529	0	0	0	1	46	154	201	110	18	0	0	0		
	CDD18.3	23	0	0	0	0	0	9	11	3	0	0	0	0		
	CDH23.3	417	0	0	0	0	32	158	164	63	0	0	0	0		
	CDH26.7	69	0	0	0	0	2	31	25	12	0	0	0	0		
	WSAvg	1.7	1.0	1.2	1.8	2.4	2.5	2.3	2.0	1.9	1.9	1.5	1.2	1.1		
(15) Precipitation	PrecAvg	319	11	9	7	7	19	52	74	64	36	21	13	14		
	PrecMax	476	41	38	35	58	57	90	196	143	66	42	43	58		
	PrecMin	243	1	0	0	0	5	19	18	12	9	5	0	1		
	PrecStd	65	10	9	7	12	13	19	45	29	17	10	11	11		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	4.8	10.0	18.4	26.6	29.5	29.1	28.1	21.1	13.0	5.2	5.1		
		MCWB	-0.3	1.0	4.2	8.8	13.5	16.4	17.5	17.5	12.6	6.4	1.3	1.4		
	2%	DB	0.0	1.4	6.9	15.4	23.8	27.3	27.0	25.0	18.9	9.5	0.5	0.6		
		MCWB	-1.7	-1.1	2.1	7.0	12.5	15.3	16.6	16.6	11.3	4.9	-0.6	-1.4		
	5%	DB	-4.2	-2.0	4.3	13.0	21.3	25.0	25.1	22.2	17.3	6.6	-2.6	-4.1		
		MCWB	-4.9	-3.2	0.5	5.8	11.2	15.0	16.2	15.0	10.5	3.5	-3.4	-4.8		
	10%	DB	-8.0	-4.9	2.0	11.2	19.0	23.0	23.3	20.0	15.4	4.0	-4.9	-7.9		
		MCWB	-8.6	-5.7	-1.1	4.8	10.1	14.0	15.6	13.9	10.1	2.1	-5.4	-8.3		
	0.4%	WB	0.5	1.3	4.4	9.1	14.1	18.1	19.5	19.2	13.4	7.0	1.8	1.9		
		MCDB	2.3	4.3	9.2	17.5	24.1	26.9	25.2	26.1	18.6	11.9	4.7	4.5		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.8	-0.8	2.3	7.6	13.2	16.7	17.9	17.1	12.1	5.4	-0.6	-1.2		
		MCDB	0.1	1.4	6.3	14.9	22.7	24.2	24.7	23.1	17.1	8.7	0.8	1.0		
	5%	WB	-4.5	-3.0	0.8	6.2	11.9	15.6	16.9	15.7	11.3	3.8	-3.2	-4.5		
		MCDB	-3.6	-1.7	4.3	12.7	19.8	22.9	23.4	20.3	15.9	6.4	-2.5	-3.7		
	10%	WB	-8.2	-5.6	-1.1	5.0	10.6	14.7	16.1	14.7	10.4	2.2	-5.2	-8.1		
		MCDB	-7.7	-4.8	1.7	10.8	17.9	21.5	22.1	18.9	14.5	4.0	-4.6	-7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	11.2	15.0	12.9	13.5	11.7	10.5	10.3	10.9	8.8	8.2	8.3		
		MCDBR	12.7	12.7	15.7	15.8	17.4	15.5	14.6	14.5	14.8	12.4	9.3	13.2		
		MCWBR	11.0	10.8	11.5	8.8	8.0	6.5	6.6	7.0	8.4	8.4	8.1	11.7		
		MCDBR	12.3	12.1	14.4	15.1	15.5	13.5	12.8	11.9	12.8	11.2	9.3	13.1		
	5% WB	MCWBR	10.7	10.3	10.5	8.6	7.2	6.0	6.4	6.1	7.6	7.7	8.1	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.173	0.249	0.255	0.319	0.344	0.382	0.375	0.395	0.307	0.263	N/A	0.178		
	taud		1.949	2.202	2.300	2.261	2.429	2.332	2.380	2.310	2.550	2.462	N/A	1.941		
	Ebn at Noon		393	671	849	852	866	831	827	762	791	674	N/A	126		
	Edh at Noon		23	54	77	103	97	110	102	100	63	43	N/A	8		
(44) All-Sky Solar Radiation	RadAvg		0.17	0.86	2.65	4.54	5.21	5.48	4.76	3.66	2.31	0.96	0.24	0.05		
	RadStd		0.01	0.07	0.17	0.29	0.39	0.45	0.33	0.36	0.27	0.11	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (5 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	-326	-354	N/A	N/A

Nomenclature: See separate page